Bias terms and decision boundaries

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make the algebra and graph agree.

Model both sides and equality,keep units and axes visible,and distinguish hand changes from evaluated improvement.

Keep one habit: Assess the new model's scores,classes,line and changed-bias result with the checked key.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Calculate a weighted score with a **bias term**

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

The contributions are 6 and -5, giving dot product 1.

Opening retrieval

Without opening the old lesson, explain one key idea from Dot products and weighted sums. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a dot product by matching feature positions

Reference example, not the only acceptable wording: Teacher explanation: Contributions are 6 and -5, so the score is 1.

Pictured end goal

By the end, I can explain **bias term**s and decision boundaries using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Dot products and weighted sums

For the compact alternative schedule, use two 45-minute sessions to connect signed arithmetic, linear equations and graph regions. Require the explicit equality policy each time.

The classroom formula is hand-selected; it demonstrates **prediction** and parameter effects, not **evidence** of successful **training**.

Outcomes

Calculate a weighted score with a **bias term**

Locate and test a two-feature **decision boundary**

Explain how changing the bias shifts a boundary without **training**
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Model algebra,axes and equality together

> Keep numeric scores separate from unsupported probability and improvement claims.

Main formula,side checks and full derivation

Use the fictional parcel schema [width in cm, height in cm]. Our model computes z = 2 times width - height - 3. The weights are [2, -1]; the bias b is -3. Compute the weighted sum first, then add the bias once.

The result is an affine score: a weighted sum plus a constant offset.

Now declare a classification rule: predict class 1 when z is at least 0, and class 0 when z is below 0. Equality belongs to class 1. These are invented categories for arithmetic practice, not a real parcel decision.

The weights and bias are supplied, so this is a forward calculation, not **training**.

A bias is a mathematical parameter. It is not the same use of the word as unfair treatment; numerical offset alone does not establish whether a system is fair.

Set z to zero to locate the boundary: 2 times width - height - 3 = 0, so height = 2 times width - 3. Plot width on the horizontal axis and height on the vertical axis. Boundary points include (2, 1), (3, 3) and (4, 5).

Join them with a straight line and label the axes in cm.

At width 3, height 2 gives score 1 and class 1; height 3 gives score 0 and class 1; height 4 gives score -1 and class 0. Under this axis convention class 1 lies on or below the boundary.

One point on each side checks that orientation.

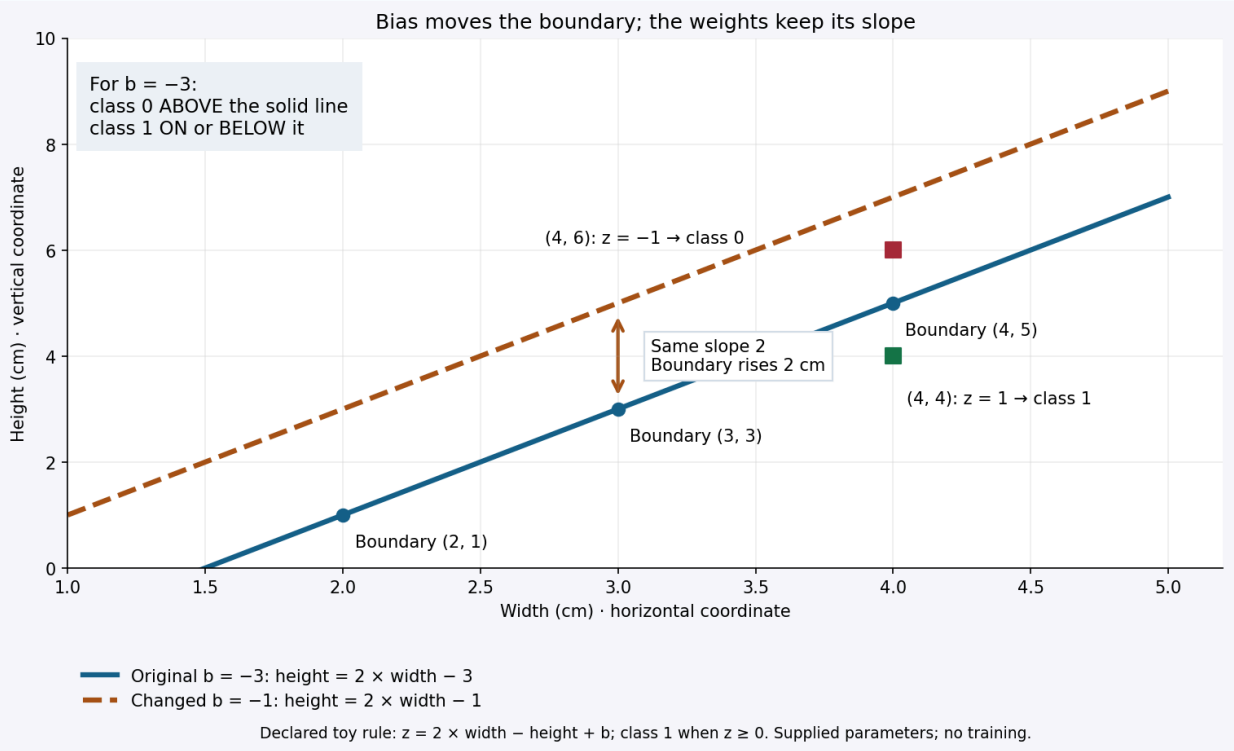
The code assumes valid numerical measurements in the agreed units. Changing >= to > changes the equality case, even though the drawn line remains in the same place.

From score to plotted boundary	Deciding step
Start with the declared score	$z=2\times\text{width}-\text{height}-3$
Locate score zero	$0=2\times\text{width}-\text{height}-3$
Isolate the vertical coordinate	$\text{height}=2\times\text{width}-3$
Plot complete boundary points	(2, 1), (3, 3), (4, 5), axes in cm
Check below and above at width 4	height 4:z1, class 1; height 6:z−1, class 0
Check equality explicitly	height 5:z0, class 1 because $z\geq 0$

Guided [3, 4]: with b = -3, z = -1 and class 0; with b = -1, z = 1 and class 1. Only b changed. Exit: test labels influenced the new bias, so the reused cases are development **evidence** for the revision.

Preserve the original result and evaluate a frozen revision on fresh independent cases.

Labelled original and shifted graph



Pictured A/Bfacts and complete scope

Move a toy boundary with a bias while keeping its **input** weights fixed.

Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

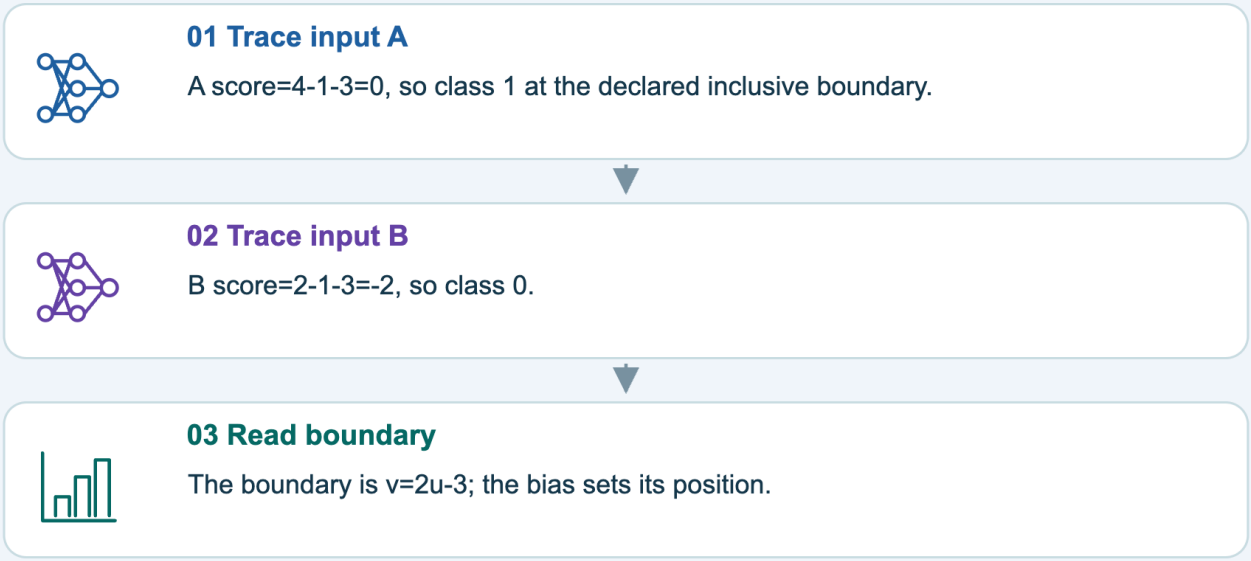
The picture uses the same formula z=2u−v+b with b=−3. A[2, 1]scores 0, class 1 at equality; B[1, 1]scores−2, class 0. Change only b to−1:both scores rise 2, A→2 and B→0, bothclass 1. The boundary changes from v=2u−3 to v=2u−1; its slope stays 2. The plotted parcel example uses width/height names for the same coordinate roles.

The hand-chosen shift is a different supplied model, not a learned improvement or probability estimate.

Trace the two original and changed scores

AI ACADEMY / AI-5/3 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Trace input A	A score=4-1-3=0, so class 1 at the declared inclusive boundary.
2. Trace input B	B score=2-1-3=-2, so class 0.
3. Read boundary	The boundary is v=2u-3; the bias sets its position.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line’s orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary.

A linear boundary is a toy classifier rule. Raw score size does not automatically describe confidence or probability.

Change: Change bias from -3 to-1.

Prediction: A score becomes 2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds 2 to every score without changing the weights.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Bias term	An additive parameter applied after a weighted sum
Decision boundary	Input locations where the declared classification rule changes class
Affine function	A weighted sum plus a constant offset
Threshold rule	A stated comparison that converts a score into a class

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
Increase b by 2	Every score rises by 2	Slope remains 2

The boundary moves parallel to itself because the weights stay fixed. An increased score does not always change a class: only inputs crossing the declared threshold change.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Hold the weights fixed while shifting the boundary

BIAS	SCORE AT [3, 3]	BOUNDARY EQUATION
b is -3	6 - 3 - 3 gives 0	height = 2 width - 3
b is -1	6 - 3 - 1 gives 2	height = 2 width - 1
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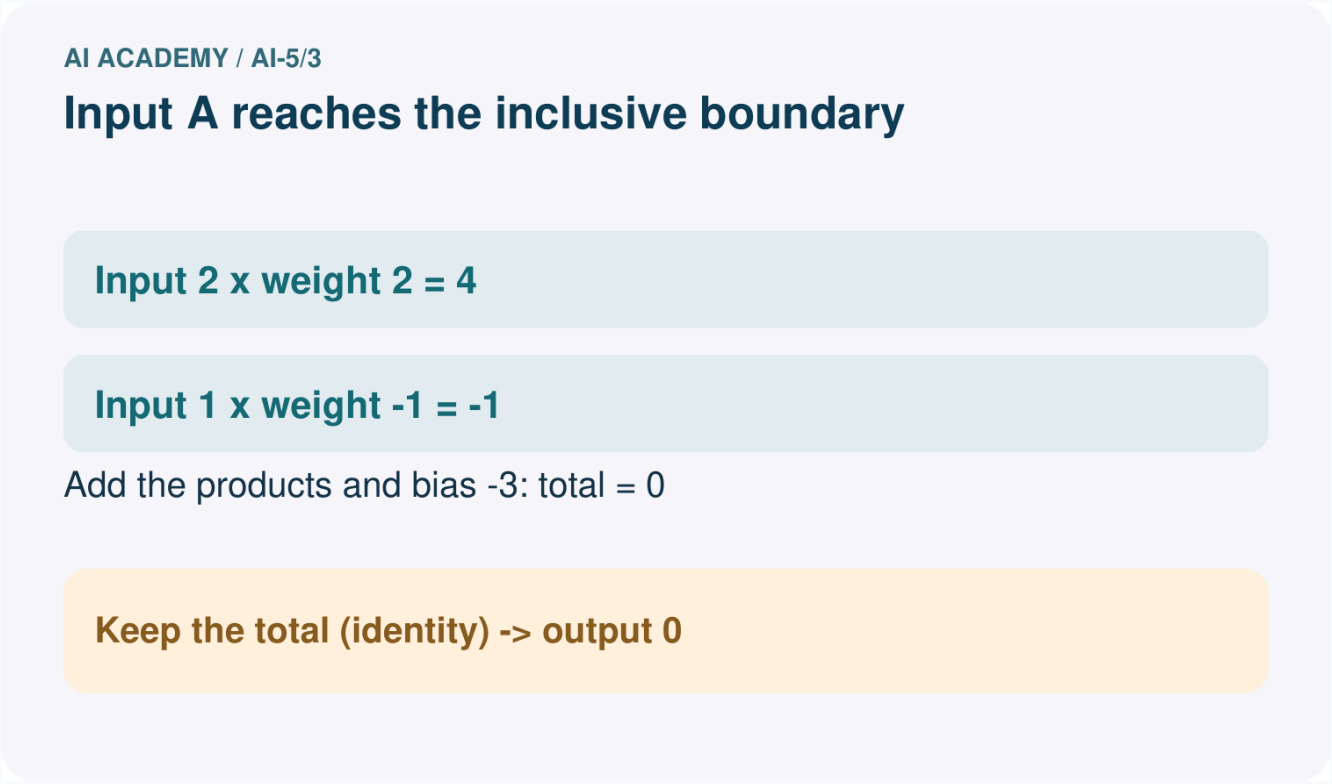
For [3, 4], compare z and prediction with bias -3 and bias -1. Explain which parameter changed.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Bias terms and decision boundaries

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/3

Bias terms and decision boundaries

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Trace input A

A score=4-1-3=0, so class 1 at the declared inclusive boundary.



2. Trace input B

B score=2-1-3=-2, so class 0.



3. Read boundary

The boundary is v=2u-3; the bias sets its position.

CHANGE ONE THING

Change bias from -3 to-1.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

Teacher check: Teacher explanation: Boundary points (2, 1) and (4, 5). [4, 4] has z = 1 and class 1; [4, 6] has z = -1 and class 0. With height vertical, class 1 is on or below the line.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Bias term**s and decision boundaries. Proposed explanation: “Add bias to each contribution”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Compute the dot product, then add one b. Adding b twice implements a different formula.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Originally B has score -2 and class 0; with bias -1 its score is 0 and class 1 because the boundary is inclusive. Changed-case check: A score becomes 2 and B score 0; both predict class 1.

Increasing bias by2 adds 2 to every score without changing the weights.

Changed-case reasoning: A score becomes 2 and B score 0; both predict class 1. Increasing bias by2 adds 2 to every score without changing the weights.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide graph paper and a partially labelled contribution table. If rearranging equations is difficult, find three zero-score points by substitution first.

Extension: multiply all weights and b by positive 2 and explain why scores double while classifications under >= 0 stay the same.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the dot product in the prerequisite question and review adding a negative constant.
7 to 18 minutes	Trace all three table rows. Students circle the one addition of b and identify the equality case.
18 to 32 minutes	Complete W1 and derive height = 2 width - 3. Plot the three boundary points on labelled axes.
32 to 45 minutes	Use W2 and the guided bias comparison. Check both sides numerically before shading a class region.
Session 2 start to 8 minutes	Retrieve the rule z >= 0. Run or trace the optional code and compare it with the plotted points.
8 to 20 minutes	Complete W3. Distinguish moving the boundary from changing the class assigned at equality; discuss the extended origin.
20 to 35 minutes	Collect W4 independently. Check scores, classes, rearranged equation and the effect of changing only b.
35 to 45 minutes	Complete W5 and the exit response. Offer the fresh retry after feedback to assess the same ideas with a new boundary.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main W2 boundary(2, 1)/(4, 5); side checks 1/−1. Equality[3, 3]z0 class1 under≥, class 0 under>; same zero line. Bias shift−3→−1 adds2 and keeps slope 2. Missing/incorrect units require schema repair. Independent w[−1, 2], b−2:inputs[2, 1]/[2, 2]/[2, 3]scores−2/0/2, classes 0/1/1, boundaryv=u/2+1. With b0, [2, 1]z0 class1, boundaryv=u/2. No final labels inform tuning.

Diagnose the current misconceptions

Add bias to each contribution: Compute the dot product, then add one b. Adding b twice implements a different formula.

Higher scores mean more correct: Correctness also needs target labels. A shifted boundary can trade one error type for another.

The **bias term** measures unfairness: This b is an additive parameter. Evaluating unfairness requires context and **evidence** about outcomes and use.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u.

Then change only b to 0 and recompute [2, 1].

Teacher explanation: Scores -2, 0, 2 give classes 0, 1, 1. Boundary -u + 2v - 2 = 0 gives v = u/2 + 1. With b = 0, [2, 1] scores 0 and class 1. The score rose 2 and the boundary moved to v = u/2.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the bias once

Use z = 2 times width - height - 3 at [5, 4]. Give the weighted sum, bias, final score and **prediction** under z >= 0.

Teacher explanation: Weighted sum 10 - 4 = 6; bias -3; score 3; **prediction** 1. Add the bias once after summing.

W2 Plot and label the boundary

For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

Teacher explanation: Boundary points (2, 1) and (4, 5). [4, 4] has z = 1 and class 1; [4, 6] has z = -1 and class 0. With height vertical, class 1 is on or below the line.

W3 Check the equality rule

At [3, 3], compare **prediction** under z >= 0 with **prediction** under z > 0. Does changing this comparison move the zero line?

Teacher explanation: Score is zero. The >= rule predicts 1; the > rule predicts 0. The zero line remains height = 2 width - 3, but its assigned class changes.

W5 Require evidence of improvement

A teammate says that raising b must improve the model because all scores rise. What information is needed before accepting an improvement claim?

Teacher explanation: Higher scores can fix some missed class -1 cases while creating false alarms.

Compare with observed labels on the same valid evaluation set under a declared metric, keeping **fitting** and model choices separate from final testing.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided [3, 4]: with b = -3, z = -1 and class 0; with b = -1, z = 1 and class 1. Only b changed. Exit: test labels influenced the new bias, so the reused cases are development **evidence** for the revision.

Preserve the original result and evaluate a frozen revision on fresh independent cases.

W1 Trace the bias once

W2 Plot and label the boundary

W3 Check the equality rule

W4 Analyse a new model independently

A new model has w = [-1, 2], b = -2, **input**s [u, v] and class 1 when z >= 0. Calculate scores and classes for [2, 1], [2, 2], [2, 3]. Write the boundary as v in terms of u. Then change only b to 0 and recompute [2, 1].

Teacher explanation: Scores -2, 0, 2 give classes 0, 1, 1. Boundary -u + 2v - 2 = 0 gives v = u/2 + 1. With b = 0, [2, 1] scores 0 and class 1. The score rose 2 and the boundary moved to v = u/2.

W5 Require evidence of improvement

Independent assessment and fresh retry

W4 should include all three scores and the equality class, with v = u/2 + 1 before the change. Fresh retry: z = u + v - 5, class 1 if z >= 0. [2, 2], [2, 3], [2, 4] give -1, 0, 1 and classes 0, 1, 1; boundary v = 5 - u.

With b changed from -5 to -4, [2, 2] reaches zero and class 1.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **bias term**s and decision boundaries **evidence** sheet to Image-pattern Network Investigation. Show how you can calculate a weighted score with a **bias term**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide graph paper and a partially labelled contribution table. If rearranging equations is difficult, find three zero-score points by substitution first.

Extension: multiply all weights and b by positive 2 and explain why scores double while classifications under >= 0 stay the same.

Week 4 compares identity, ReLU and sigmoid transforms of the score.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/4, ai-5/6, ai-5/9. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Calculate a weighted score with a **bias term**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For the worked model, find boundary points at width 2 and width 4. Test [4, 4] and [4, 6] to identify the class on each side.

Reasoned practice answer: Teacher explanation: Boundary points (2, 1) and (4, 5). [4, 4] has z = 1 and class 1; [4, 6] has z = -1 and class 0. With height vertical, class 1 is on or below the line.

Check every learner before discussion: Use the worked case for **Bias term**s and decision boundaries. Proposed explanation: “Add bias to each contribution”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Compute the dot product, then add one b. Adding b twice implements a different formula.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Change bias from -3 to-1. Require a **prediction** and a reason before comparing. Expected result: A score becomes 2 and B score 0; both predict class 1.

Reason: Increasing bias by2 adds 2 to every score without changing the weights.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

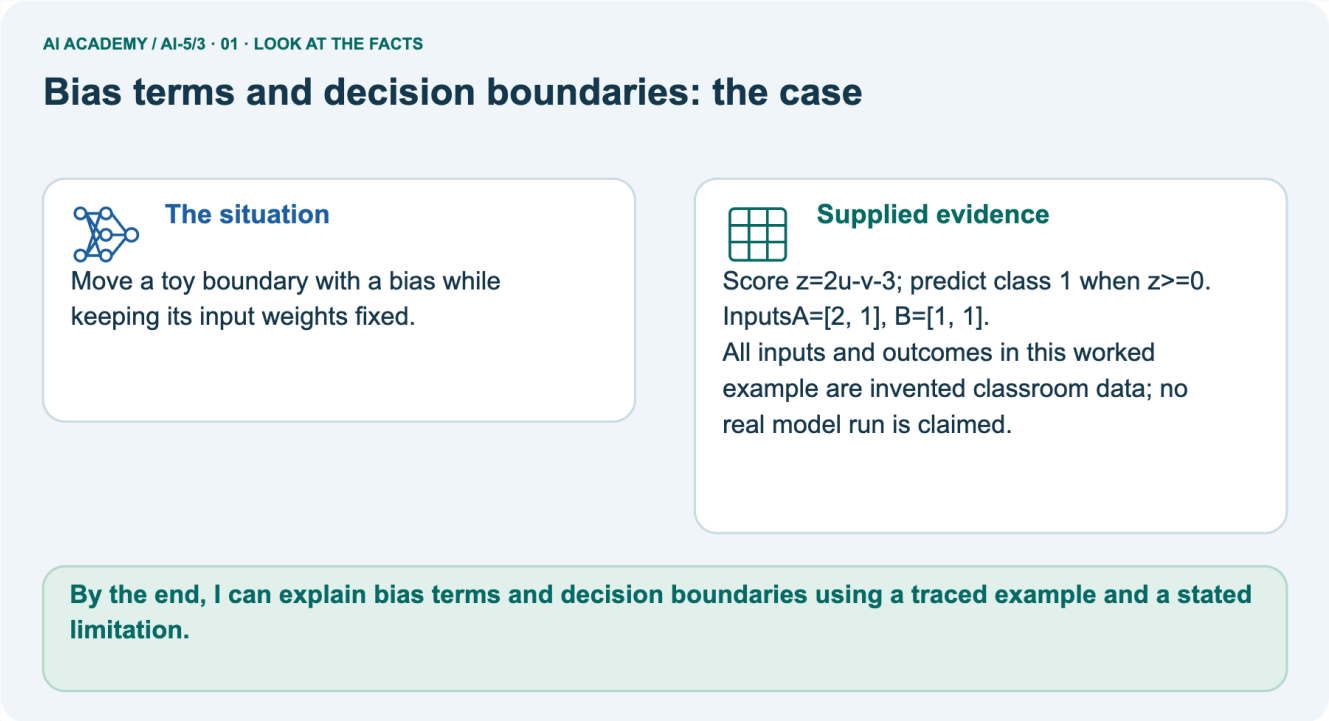
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Bias terms and decision boundaries: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

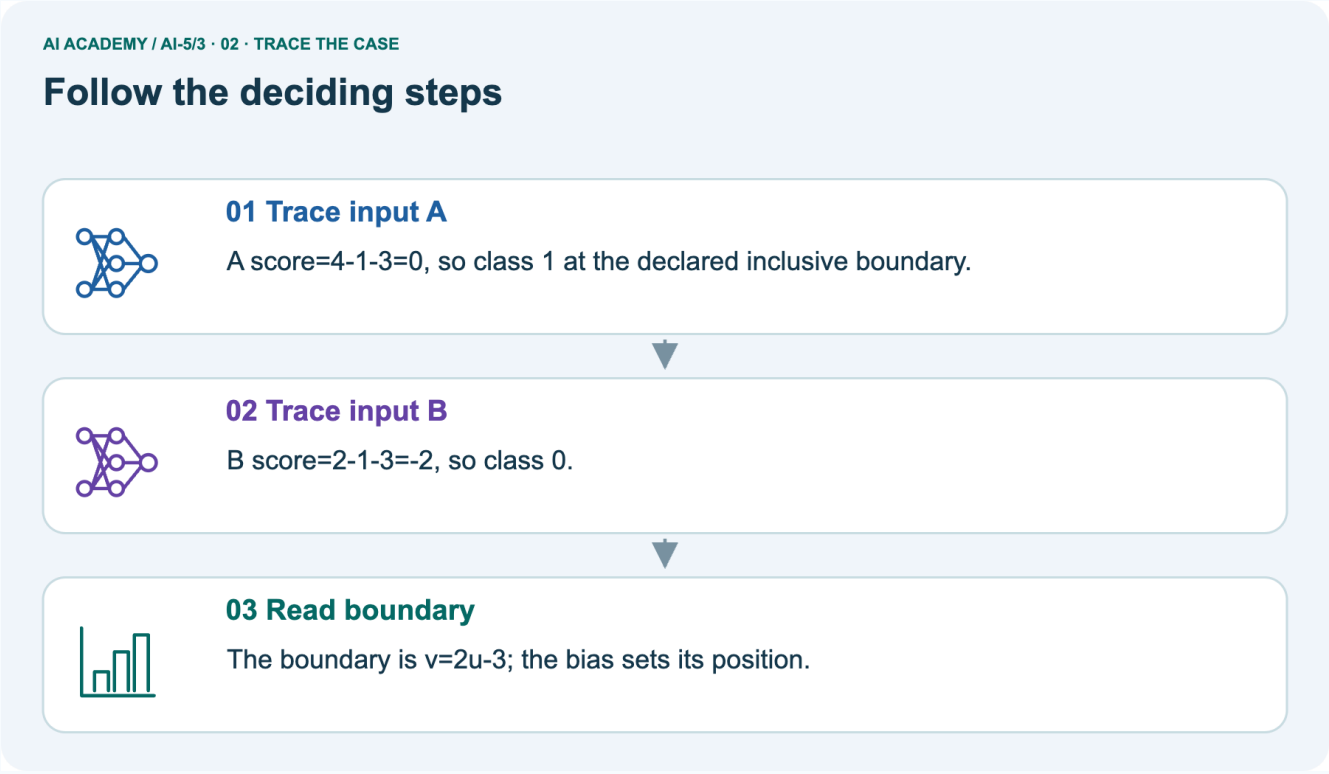
Move a toy boundary with a bias while keeping its **input** weights fixed. Supplied facts: Score z=2u-v-3; predict class 1 when z>=0. **Input**sA=[2, 1], B=[1, 1].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



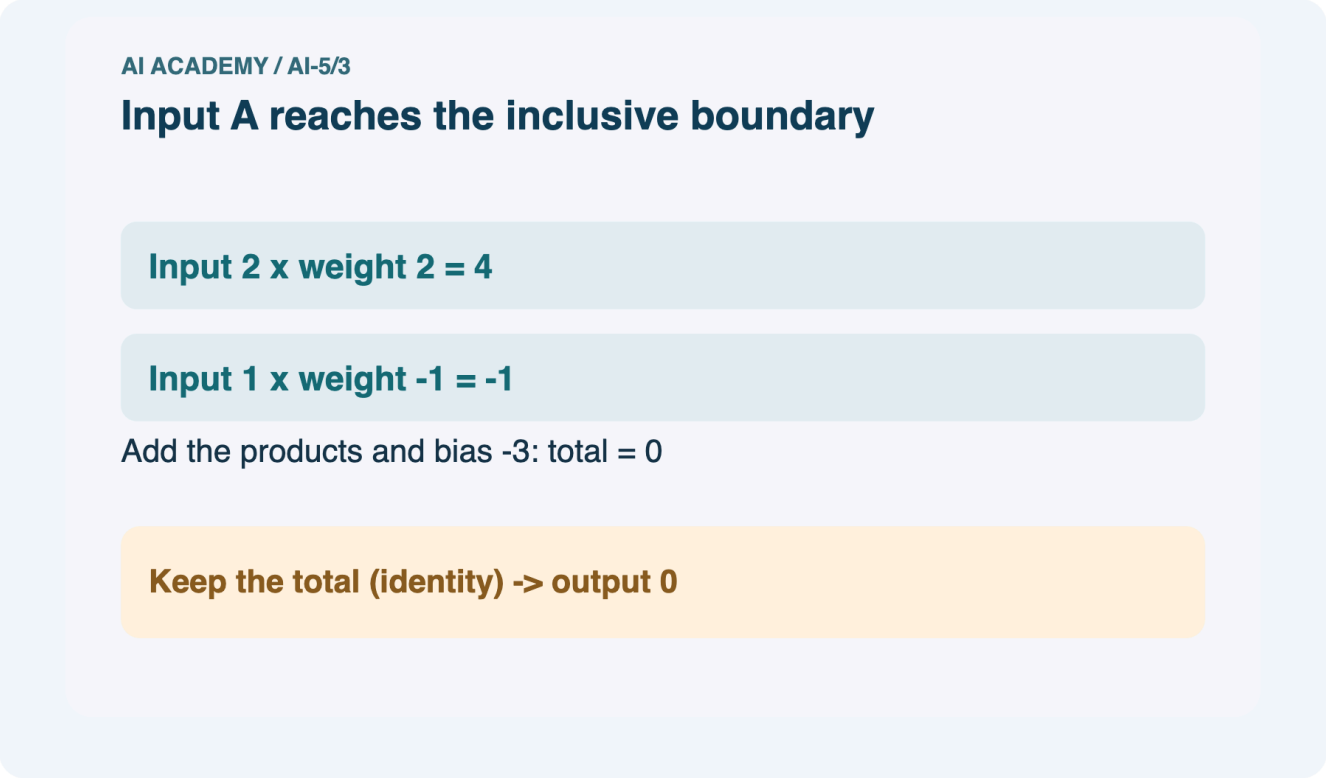
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace **input** A: A score=4-1-3=0, so class 1 at the declared inclusive boundary. → Trace **input** B: B score=2-1-3=-2, so class 0. → Read boundary: The boundary is v=2u-3; the bias sets its position.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Input A reaches the inclusive boundary



Follow the labelled picture with the matching explanation. Open the vector diagram

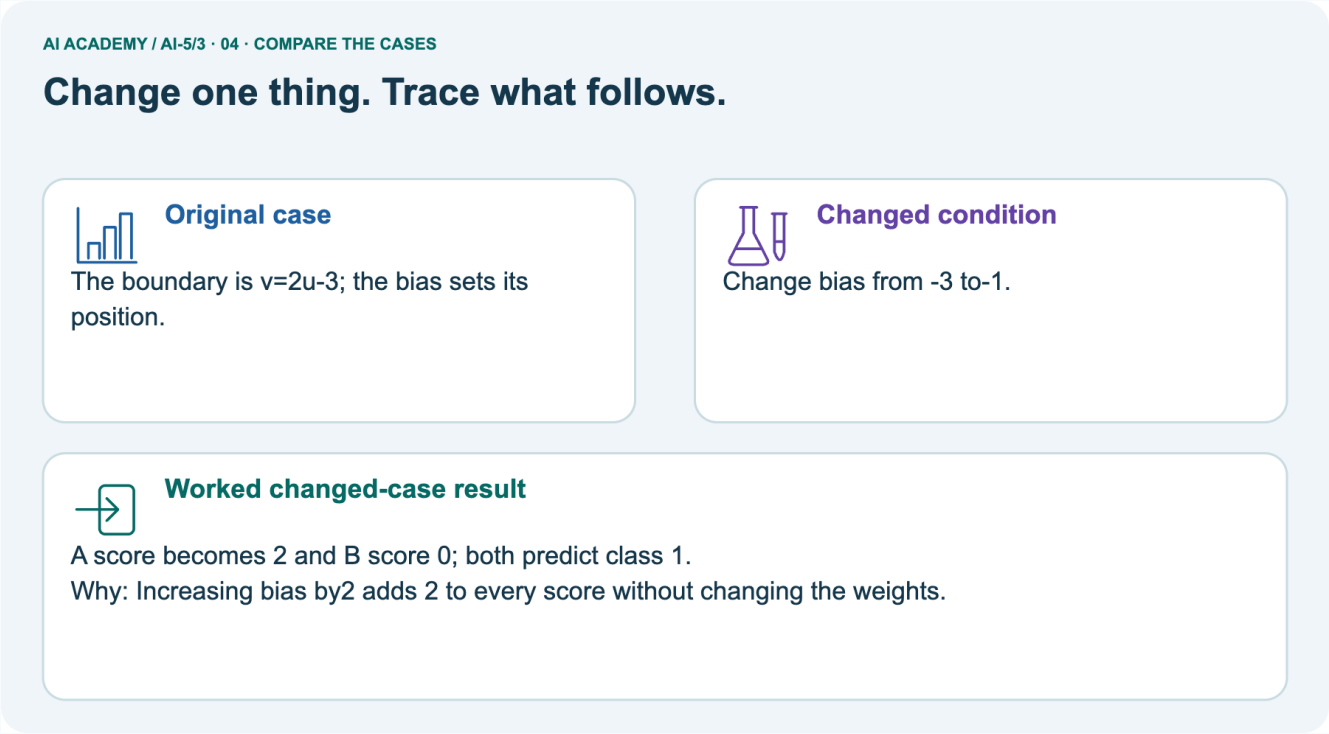
Input A reaches the inclusive boundary. **Input**s: [2, 1]; weights: [2, -1]; bias: -3; activation: identity; **output**: 0.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The boundary is v=2u-3; the bias sets its position. Changed condition: Change bias from -3 to-1. Expected result: A score becomes 2 and B score 0; both predict class 1.

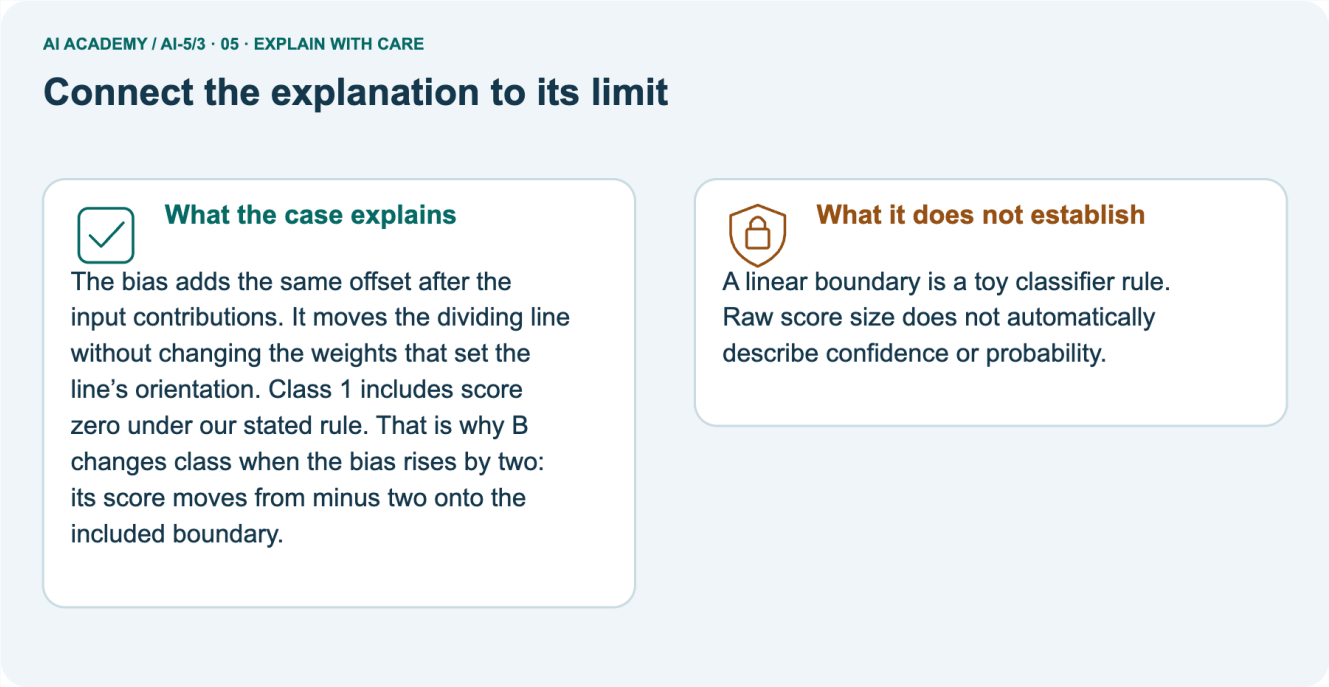
Reason: Increasing bias by2 adds 2 to every score without changing the weights.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Increasing bias by2 adds 2 to every score without changing the weights.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The bias adds the same offset after the **input** contributions. It moves the dividing line without changing the weights that set the line's orientation. Class 1 includes score zero under our stated rule.

That is why B changes class when the bias rises by two: its score moves from minus two onto the included boundary. Limit: A linear boundary is a toy classifier rule.

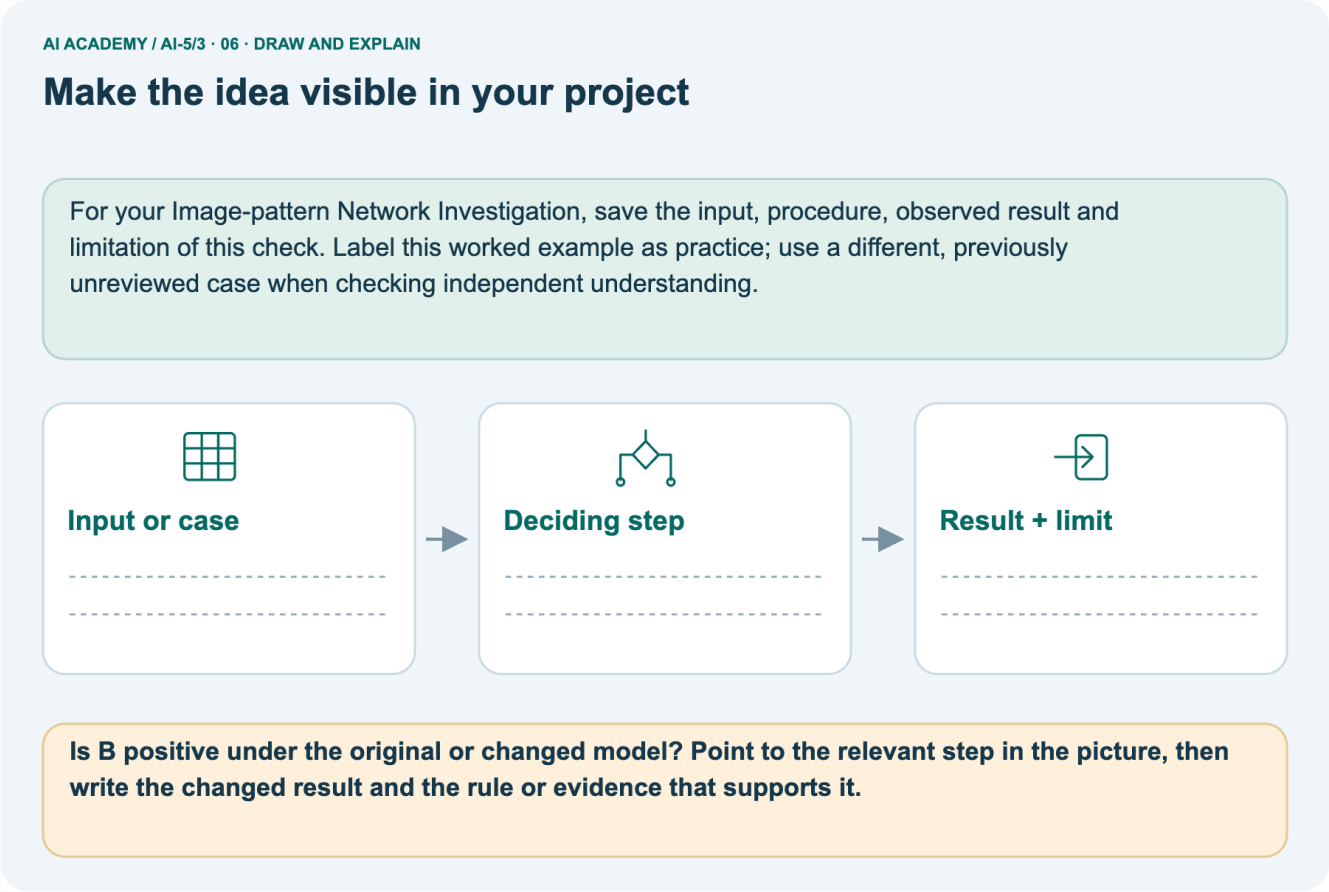
Raw score size does not automatically describe confidence or probability.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A linear boundary is a toy classifier rule. Raw score size does not automatically describe confidence or probability.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Is B positive under the original or changed model? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Originally B has score -2 and class 0; with bias -1 its score is 0 and class 1 because the boundary is inclusive. Changed-case check: A score becomes 2 and B score 0; both predict class 1.

Increasing bias by2 adds 2 to every score without changing the weights.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Bias terms and decision boundaries: the case

02. Follow the deciding steps

03. Input A reaches the inclusive boundary

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.